

“... gun that men carry... fires of its own action
... they are small... nobody sees them...”

DUCAL EDICT, BRESCIA, NORTHERN ITALY (1532)

THE WHEEL-LOCK MECHANISM

The first gunlock to overcome these problems was based on a tinder-lighter—a simple device used to kindle fire. This “wheel-lock” demanded great skill to build. It consisted of a steel wheel that rotated against a piece of iron pyrite, a natural mineral, to produce sparks. One end of the lock’s V-shaped mainspring was attached to a chain. By using a key to turn the wheel, the shooter wound this chain around the

mechanism’s axle, compressing the spring (“spanning” the lock). He then locked the spanned wheel in position, preventing it from spinning. At this point, the upper edge of the wheel entered the pan through a slot. Next, he placed gunpowder in the priming pan and closed the cover. When the gun was to be fired, the shooter moved the dog (the part of the lock that held the iron pyrite), bringing it over by hand, and placing it onto the pan cover. Pulling the trigger released the wheel, which automatically opened the pan cover. The iron pyrite hit the rotating wheel, producing sparks, which flashed through a touchhole on the side of the barrel to light the main gunpowder charge in the barrel’s breech.

NEW WEAPONS

The wheel-lock design enabled the manufacture of firearms that could be carried primed and ready to fire. Because they did not require live fire, firearms could now be carried concealed. It made a brand new kind of small firearm—the pistol—a practical proposition by the 1520s. Single-handed operation of firearms became possible. The thought of a firearm small enough to be concealed under clothing alarmed European authorities, who considered it a threat to public safety. By the early 16th century, many European countries had introduced legislation against these new, portable firearms.

Thanks to the wheel-lock’s portability, the cavalry at last had firearms that they could use effectively on horseback, without the need to dismount. Wheel-lock firearms, such as pistols and carbines (see p. 32), could be stowed away for use at a moment’s notice. Each weapon could be fired only once during an engagement, which was why cavalry were issued with pairs of pistols, and sometimes carbines too. This, however, gave them the advantage of two or indeed three shots from the saddle, when previously none had been possible. This offered the cavalry firepower like never before.

◀ SHOOTING ON HORSEBACK

During the Thirty Years’ War, at the Battle of Lützen (November 16, 1632), the Protestant Swedish king, Gustavus Adolphus, led his cavalry against Catholic Imperial forces. Shot by Imperial cavalrymen wielding wheel-lock pistols, he succumbed to his injuries.

KEY FIGURE

Leonardo da Vinci
(1452–1519)

The earliest images of a mechanism resembling a wheel-lock appeared in the notes of Leonardo da Vinci’s *Codex Atlanticus* in around 1495. It seems Leonardo was inspired by a tinder-lighter when he made drawings of a fire-striking device to attach to the side of a gun barrel.



AFTER »

Although the invention of the wheel-lock enabled the development of new handheld arms that could be portable, concealed, and used on horseback, there were still drawbacks. The wheel-lock was costly, easily put out of order, and hard to repair—problems in both military and hunting situations. A simpler, more reliable gunlock was still needed.

- **RARE LEVELS OF EXPERTISE** were needed to manufacture wheel-lock pistols, which made them expensive guns to buy.

- **THE SNAPHANCE LOCK** (see p. 38), a precursor to the flintlock, evolved in the 1560s.

FLEMISH FLINTLOCK PISTOL



- **THE FLINTLOCK** appeared during the 1570s (see pp. 38–39). It was cheaper, simpler, and more reliable than the wheel-lock or the matchlock.